Q16

The resistance should start at zero (s = 0), increase in magnitude and then end at zero (s = L).
Let resistance per unit length of the wire be k.
The resistance between X and Y will be that of two resistors in parallel.

Thus $R = \left(\frac{1}{R_1} + \frac{1}{R_2}\right)^{-1} = \frac{R_1 R_2}{R_1 + R_2}$

or $R = \frac{(ks)[k(L-s)]}{ks + k(L-s)} = \frac{ks(L-s)}{L}$

The variation of R with s is therefore not a straight-line relationship.

Ans: B

Q17